MILITARY SPECIFICATION SHEET

ANTENNA, BLADE, AT-741B/B (HN CONNECTOR)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the antenna described herein shall consist of this document and the latest issue of Specification MIL-A-25708.

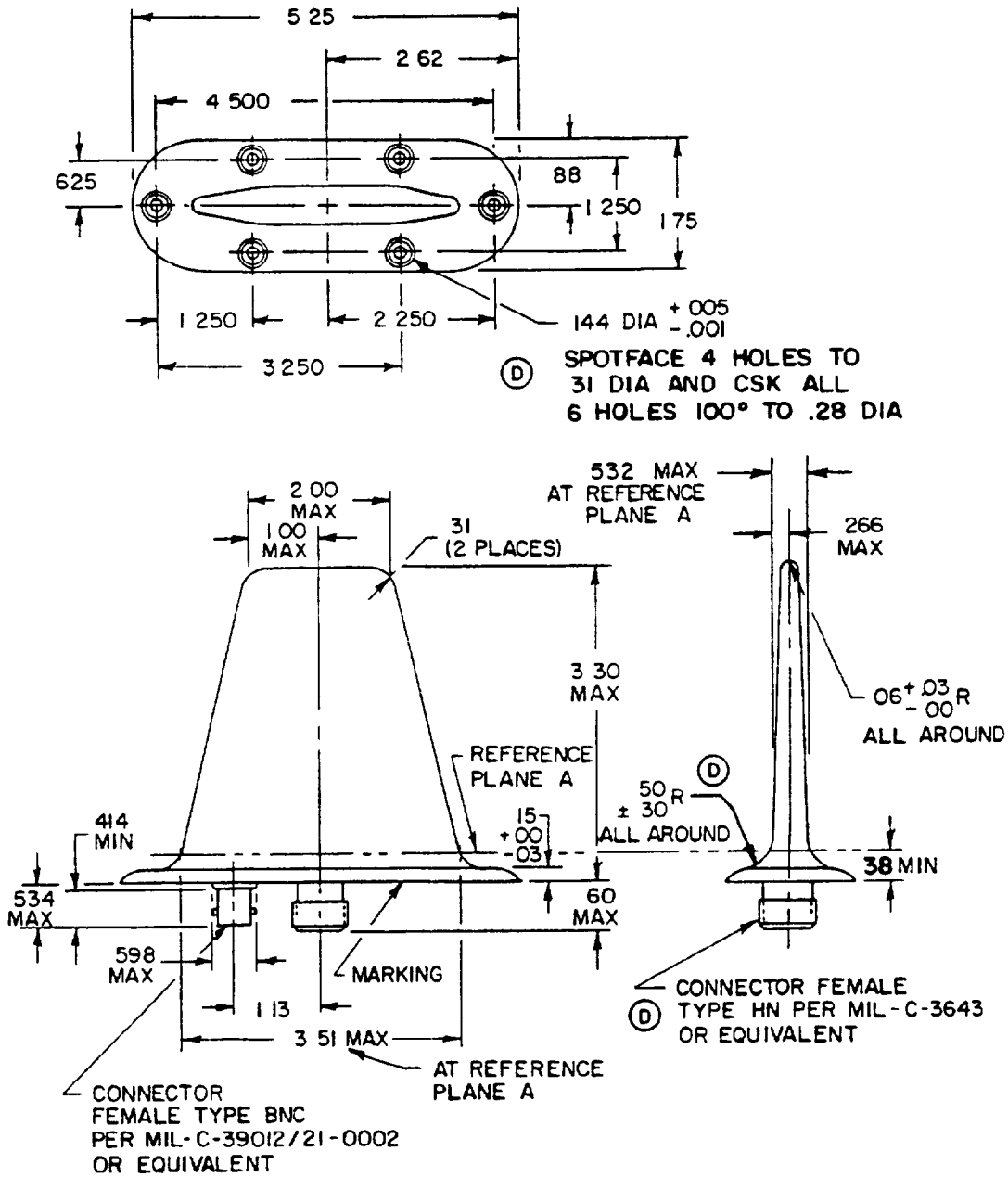


FIGURE 1.
Ⓓ denotes changes

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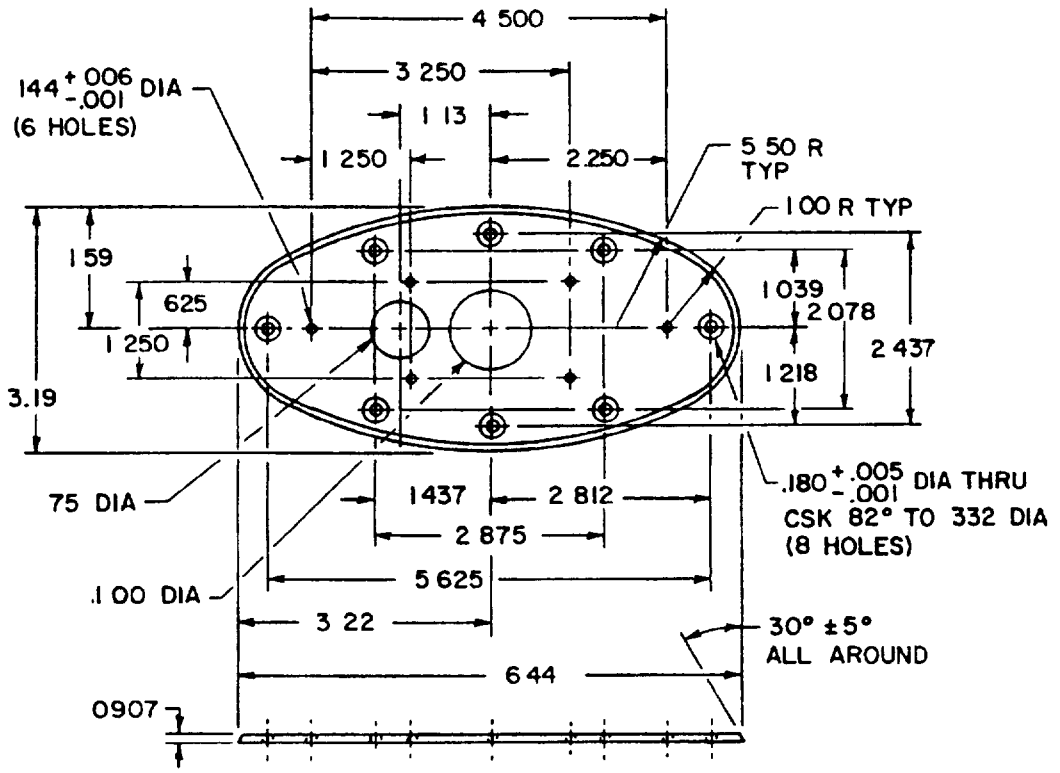
INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.02	.30	7.62	.60	15.24	2.250	57.15
.005	.13	.31	7.87	.625	15.88	2.62	66.55
.03	.76	.35	8.89	.88	22.35	3.250	82.55
.06	1.52	.414	10.52	1.00	25.40	3.30	83.82
.144	3.66	.50	12.70	1.13	28.70	3.51	89.15
.15	3.81	.532	13.51	1.250	31.75	4.500	114.30
.266	6.76	.534	13.56	1.75	44.45	5.25	133.35
.28	7.11	.598	15.19	2.00	50.80		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are $\pm$ .005 (.13 mm) for three place decimals and + .02 (.51 mm) for two place decimals.
4. The base of the antenna is considered that part that rises from the bottom of the antenna mounting surface to the reference plane "A". Must be not less than .35 (8.89 mm). The cover is that part above reference plane "A"

FIGURE 1. - Continued

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.02	.75	19.05	1.59	40.39	3.22	81.79
.005	.13	1.00	25.40	2.078	50.78	3.250	82 55
.0907	2.30	1.039	26.39	2.250	57.15	4.500	114.30
.144	3.66	1.13	28.70	2.437	61.90	5.50	139.70
.180	4.57	1.218	30.94	2.812	71.42	5.625	142.88
.332	8.43	1.250	31.75	2.875	73.02		
.625	15.88	1.437	36.50	3.19	81.03		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ± .005 (.13 mm) for three place decimals and ± .01 (.25 mm) for two place decimals.

FIGURE 2. Adapter plate P/N M25708/1-02.

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REQUIREMENTS (ANTENNA):

Dimensions and configuration: See figure 1.

Materials:

Base : Aluminum alloy per QQ-A-591, alloy 380.
Cover: Suitable material that will resist errosion of impinging particles at velocities up to Mach 2.

Temperature operating range: -55°C to 250°C.

(D) Static load 15 PSIG

Frequency range: .960 to 1.220 GHz.

Impedance: 50 ohms nominal.

VSWR:

Antenna	1.40:1 maximum from .960 to 1.220 GHz.
	1.30.1 maximum from 1.000 to 1 100 GHz
Sampling probe:	1.50.1 maximum from .960 to 1.220 GHz.

Power : 4kW peak, 100 watts average.

Weight. 10 ounces maximum

Part Number. See table I.

Data list: A data list shall accompany each antenna with VSWR measurement values made at .960, 1.000, 1.100, and 1.200 GHz.

REQUIREMENTS (ADAPTER PLATE):

Dimensions and configuration. See figure 2.

Material: Aluminum alloy per QQ-A-250/11, Temp T4.

Finish. Anodize per MIL-A-8625, Type I, Class I.

TABLE I. PART Number

Part Number	Description
M25708/1-01	AT-741B/B Antenna
1/ MS25708/1-02	AT-741B/B Adapter

1/ Inactive for new design. To be used to replace existing AS-133/APX and AT-722/APX antennas by using antenna AT-741B/B (M25708/1-01) and antenna adapter plate (M25708/1-02).

TABLE II. Supersession data

Part Number	Replaces
M25708/1-01	AT-741B/A
M25708/1-01	AT-741A/A
M25708/1-01	AT-741/A
M25708/1-01 and M25708/1-02	AS-133/APX
M25708/1-01 and M25708/1-02	AT-722/APX

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Custodians
Army - ER
Navy - AS
Air Force - 85

Review activities
Army - AV
Air Force - 11, 80
DLA - ES

User activities
Air Force - 70, 71, 84

Preparing activity
Air Force - 85

(Project 5985-1006)

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